

Security Assessment Report

Confidential

Engagement: Network Traffic Analysis

Scope: cap3.pcapng traffic capture analysis

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1. Executive Summary

During the assessment of the provided network traffic capture, a critical security misconfiguration was identified. A private RSA key was exposed via an unsecured HTTP endpoint. Due to the use of TLS_RSA without Perfect Forward Secrecy (PFS), full TLS session decryption was successfully demonstrated.

Overall Risk Rating: CRITICAL

2. Scope and Methodology

- 1 Passive analysis of cap3.pcapng network traffic
- 2 Identification of cryptographic weaknesses
- 3 Validation of exploitation feasibility
- 4 Assessment aligned with CVSS v3.1 scoring methodology

3. Findings Overview

ID	Finding	Severity	CVSS v3.1
F-01	Private Key Disclosure	Critical	9.8
F-02	Lack of Perfect Forward Secrecy	Critical	9.1
F-03	Deprecated TLS 1.0 Usage	High	7.4

4. Detailed Finding – F-01 Private Key Disclosure

Description: The endpoint /k3y exposed a private RSA key in PEM format over HTTP without authentication.

Impact: An attacker can decrypt TLS sessions, impersonate the server, and perform man-in-the-middle attacks resulting in complete confidentiality compromise.

Proof of Exploitation: The extracted key was imported into Wireshark, enabling successful TLS session decryption and retrieval of protected resource 'file'.

Recommendation: Immediately revoke the exposed private key, reissue certificates, enforce TLS 1.2/1.3, and enable ECDHE cipher suites.